



Computational Chemistry Comparison and Benchmark DataBase

Release 22 (May 2022)

Standard Reference Database 101 [National Institute of Standards and Technology](https://www.nist.gov)

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Experimental data for CN (Cyano radical)

22 02 02 11 45

Other names
Cyano radical;

INChI	INChIKey	SMILES	IUPAC name
InChI=1S/CN/c1-2	JEVCWSUVFOYBFI-UHFFFAOYSA-N	[C]#N	

State	Conformation
2Σ	C*V

Enthalpy of formation (Hfg), Entropy, Integrated heat capacity (0 K to 298.15 K) (HH), Heat Capacity (Cp)

Property	Value	Uncertainty	units	Reference	Comment
Hfg(298.15K) $\Delta_f H^\circ$	440.29	5.00	kJ mol ⁻¹	Gurvich	
Hfg(0K) $\Delta_f H^\circ$	437.00	5.00	kJ mol ⁻¹	Gurvich	
Entropy (298.15K) S°	202.64		J K ⁻¹ mol ⁻¹	Gurvich	
Integrated Heat Capacity (0 to 298.15K) $H^\circ - H^\circ(T)$	8.67		kJ mol ⁻¹	Gurvich	
Heat Capacity (298.15K) C_p°	29.16		J K ⁻¹ mol ⁻¹	Gurvich	

Information can also be found for this species in the [NIST Chemistry Webbook](#)

Vibrational levels (cm⁻¹)

Mode Number	Symmetry	Frequency			Intensity			Comment	Description
		Fundamental(cm ⁻¹)	Harmonic(cm ⁻¹)	Reference	(km mol ⁻¹)	unc.	Reference		
1	Σ	2042	2069	1979HUB/HER					

Detailed diatomic data

ω_e	$\omega_e x_e$	$\omega_e y_e$	B_e	α_e	ZPE	reference
2068.648	13.0971	-0.0124	1.899783	0.017372	1031.133	2007Iri:389

vibrational zero-point energy: 1021.2 cm⁻¹ (from fundamental vibrations)[Calculated vibrational frequencies](#) for CN (Cyano radical).

More spectroscopic constants are available at the NIST Physics Laboratory website:

<http://physics.nist.gov/PhysRefData/MolSpec/Diatomic/Html/Tables/CN.html>

Rotational Constants (cm⁻¹)

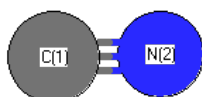
See section [I.F.4 to change rotational constant units](#)

A	B	C	reference	comment
	1.89974		1979HUB/HER	

[Calculated rotational constants](#) for CN (Cyano radical).

Product of moments of inertia $I_A I_B I_C$			
8.873652	amu Å ²	1.473527E-39	gm cm ²

Geometric Data

**Point Group $C_{\infty v}$** **Internal coordinates**

distances (r) in Å, angles (a) in degrees, dihedrals (d) in degrees

Description	Value	unc.	Connectivity				Reference	Comment
			Atom 1	Atom 2	Atom 3	Atom 4		
rCN	1.172		1	2			1979HUB/HER	re

Cartesians

Atom	x (Å)	y (Å)	z (Å)
C1	0.0000	0.0000	0.0000
N2	0.0000	0.0000	1.1718

Atom - Atom Distances

Distances in Å

	C1	N2
C1		1.1718
N2	1.1718	

[Calculated geometries](#) for CN (Cyano radical).**Bond descriptions**

Examples: C-C single bond, C=C, double bond, C#C triple bond, C:C aromatic bond

Bond Type	Count
C#N	1

Connectivity

Atom 1	Atom 2
C1	N2

Electronic energy levels (cm⁻¹)

Energy (cm ⁻¹)	Degeneracy	reference	description
0	2	1979HUB/HER	$^2\Sigma$
9245.28	4	1979HUB/HER	
25752	2	1979HUB/HER	

Ionization Energies (eV)

Ionization Energy	I.E. unc.	vertical I.E.	v.I.E. unc.	reference
13.598				webbook

Electron Affinity (eV)

Electron Affinity	unc.	reference
3.862	0.005	webbook

Proton Affinity (kJ mol⁻¹)

Proton Affinity	unc.	Product	reference	comment
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		HCN ⁺		
		HNC ⁺		

Dipole, Quadrupole and Polarizability

Electric dipole moment $\leftrightarrow$

State	Config	State description	Conf description	Exp. min.	Dipole (Debye)				Reference	comment	Point Group	Components	
					x	y	z	total				dipole	quadrupole
1	1	$^2\Sigma$	$C_{\infty V}$	True	0.000	0.000	-1.450	1.450	1968Tho/Dal:2815		$C_{\infty V}$	1	1

Experimental dipole measurement abbreviations: MW microwave; DT Dielectric with Temperature variation; DR Indirect (usually an upper limit); MB Molecular beam

[Calculated electric dipole moments](#) for CN (Cyano radical).

Electric quadrupole moment $\leftrightarrow\leftrightarrow$

State	Config	State description	Conf description	Exp. min.	Quadrupole (D Å)			Reference	comment	Point Group	Components	
					xx	yy	zz				dipole	quadrupole
1	1	$^2\Sigma$	$C_{\infty V}$	True						$C_{\infty V}$	1	1

[Calculated electric quadrupole moments](#) for CN (Cyano radical).

References

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squib	reference	DOI
1968Tho/Dal:2815	R Thomson, FW Dalby "Experimental determination of the dipole moments of the X ($2\Sigma^+$) and B($2\Sigma^+$) state of the CN molecule" Canadian Journal of Physics 46, 2815 (1968)	10.1139/p68-652
1979HUB/HER	Huber, K.P.; Herzberg, G., Molecular Spectra and Molecular Structure. IV. Constants of Diatomic Molecules, Van Nostrand Reinhold Co., 1979	10.1007/978-1-4757-0961-2
2007Iri:389	KK Irikura "Experimental Vibrational Zero-Point Energies: Diatomic Molecules" J. Phys. Chem. Ref. Data 36(2), 389, 2007	10.1063/1.2436891
Gurvich	Gurvich, L.V.; Veyts, I. V.; Alcock, C. B., Thermodynamic Properties of Individual Substances, Fouth Edition, Hemisphere Pub. Co., New York, 1989	
webbook	NIST Chemistry Webbook (http://webbook.nist.gov/chemistry)	10.18434/T4D303

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AlN (Aluminum nitride)	CN ⁻ (cyanide anion)